TA14 AUTHORITY GOVERNANCE INSTITUTION

From Laboratory Efficacy to Authorized Deployment

Technical Reference v2.0 | Institutional Examination Edition | 15 October 2026

Independent technical examination responding to research-to-deployment challenges raised in correspondence concerning EPA–Blueprint Biosecurity CRADA work. Not an EPA or Blueprint document, endorsement, evaluation, or agreed collaboration. No confidential CRADA information was used.

Executive determination boundary

Laboratory efficacy establishes only the claim supported by a particular test method, endpoint, apparatus, and operating conditions. It does not itself establish field performance, occupant safety, reduced disease risk, regulatory approval, site permission, or authority to execute. TA14 examines the proposed consequence through a traceable evidence-and-authority procedure.

Governing question: Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

Canonical chain: Reality → Record → Continuity → Admissibility → Binding → Commit → Execution → Outcome.

Disposition vocabulary: ALLOW / HOLD / DENY / ESCALATE. A disposition applies to a defined proposal, not to an entire technology in the abstract.

1 | Scope, exclusions, and institutional interfaces

Applies to laboratory-to-field translation, air-treatment trials, occupied-space deployment, continued operation, and scale-up. The examination does not substitute for standardized scientific testing, toxicology, occupational or environmental safety review, human-subject protections, procurement, site governance, professional licensure, or regulatory determinations.

A scientific finding and an execution authorization are separate records. An independent TA14 examination may inform an authorized decision maker; it does not confer agency, school, manufacturer, or facility authority on TA14.

2 | The five stated research-to-deployment challenges

Challenge	Specific examination contribution
Standardized efficacy testing	Preserve protocol, controls, target endpoint, raw observations, uncertainty, deviations, and claim boundary.
Lab-to-field prediction	Compare test and field conditions; document assumptions and unverified transfer mechanisms.
Public confidence	Publish inspectable claims, limitations, responsibility, correction and withdrawal pathways without disclosing protected data.
Investment prioritization	Bind resource decisions to stated alternatives, evidence sufficiency, authority, and review triggers.
Regulatory frameworks	Identify applicable requirements and the competent decision maker; do not infer approval from evidence.

3 | Scientific transferability matrix

Each dimension must be marked matched, bounded difference, materially different, or unknown, with references and a rationale. “Matched” requires a defensible criterion, not visual similarity.

Dimension	Laboratory record	Field evidence needed
Target	Challenge organism, surrogate, aerosol or chemical	Target relevance, exposure pathway and endpoint mapping
Airflow	Chamber volume, air changes, mixing, placement	Ventilation, mixing, leakage, room geometry, source rates
Population	No occupants or controlled test exposure	Occupancy, vulnerable populations, time-activity
Dose and time	Concentration, treatment intensity, duration	Realistic concentration, duty cycle and exposure duration
Performance	Removal/inactivation endpoint and uncertainty	Field-relevant endpoint, controls and confounders
Safety	Measured byproducts and test limits	Secondary products, exposure limits, failure modes
Continuity	Calibrated short-duration test	Maintenance, drift, degradation, changes and monitoring

4 | Evidence package and quantitative requirements

• Claim register: exact outcome, population, site, timescale, denominator, comparator, minimum meaningful effect and prohibited extrapolations.

• Methods: protocol/version, sampling frame, controls, blanks, replicates, instrument traceability, calibration, detection limits, missing data, uncertainty intervals, and deviations.

• Transferability: predeclared comparability thresholds for ventilation, airflow, source rates, geometry, occupancy, operating conditions and target contaminants.

• Safety: foreseeable byproducts, secondary exposure pathways, vulnerable occupants, fault conditions, monitoring, stop limits and competent reviewer.

• Decision integrity: responsible actor, authority source, standing basis, expiration, operating constraints, evidence IDs and decision version.

No universal efficacy threshold is asserted here. Numeric acceptance criteria must be selected by the responsible scientific and safety experts for the specific technology and use case, recorded before the decision, and tied to applicable standards or justified study design.

Preserve raw and adjusted records separately. For each adjustment retain formula, input, version, reason, uncertainty impact, author and timestamp. Unobserved effects must not be presented as measured outcomes.

5 | Six examination gates and disposition triggers

Gate	Required examination	Typical trigger
G1 Claim	Is the proposed field claim supported by the test endpoint?	HOLD expanded claims without evidence
G2 Transfer	Are field conditions characterized and transfer assumptions supported?	HOLD material unknowns
G3 Safety	Are adverse exposures and byproducts reviewed by competent specialists?	ESCALATE unresolved specialist safety matters
G4 Continuity	Can installation, operation, maintenance, calibration and change events be traced?	HOLD broken continuity
G5 Authority	Does the actor possess applicable authority and established standing at this site?	DENY action outside authority
G6 Verification	Are success criteria, stop rules, rollback and reassessment defined?	HOLD until bounded

6 | Standing is not the same as authority

The Admissible Standing Architecture (ASA) requires four separately recorded anchors: ACTOR (who acts), CAPACITY (in what role), STANDING BASIS (what documentary or institutional basis permits involvement), and AUTHORITY RELATIONSHIP (how the actor's role connects to the actual decision). Applicable Authority and Established Standing must each be established; neither is inferred from expertise or access to data.

For example, a manufacturer may provide credible laboratory evidence without having permission to install a device in a public school. A school administrator may control site access without being competent to resolve unresolved toxicology questions.

7 | Two independent decision records

Decision A: efficacy/claim	Decision B: execution/deployment
What proposition is scientifically supported?	What action may a specific actor take at a specified site and time?
Endpoint, controls, uncertainty, transfer assumptions	Authority, standing, safety approvals, conditions and monitoring
Possible HOLD on a broad infection-risk claim	Possible separate ALLOW for a narrowly scoped, approved evaluation
Does not authorize installation or operation	Does not validate an unsupported scientific claim

These dispositions must never be collapsed into a single PASS. A field evaluation may be conditionally authorized while a broader marketing or health claim remains on HOLD; conversely, a scientifically supported claim does not automatically authorize deployment.

8 | Hypothetical classroom examination: initial record

Illustrative facts only: A vendor presents a chamber test showing a reduction in a defined airborne surrogate. A school proposes use in occupied classrooms and describes it as proven reduction of infection risk. The chamber record contains a defined setup, but the case provides no classroom mixing study, no exposure-based disease endpoint, no site safety review, and no documented school authorization.

Record	Evidence finding	Disposition
A: broad infection-risk claim	Surrogate chamber reduction does not establish real-world infection-risk reduction	HOLD
B: occupied classroom deployment	Field transfer, safety and school authorization are incomplete	HOLD
C: proposed bounded field study	Could be examined separately after protocol, ethics/safety and site permissions are established	NOT YET DETERMINED

Evidence required to reopen

• Exact laboratory report and original observations, including uncertainties and deviations.

• Classroom characterization and a predeclared comparison method.

• Technology-specific byproduct and exposure assessment, including vulnerable occupants.

• Documented school decision authority, actor capacity, standing basis and relevant regulatory interface.

• Field protocol with controls, success criteria, operating limits, adverse-event reporting and stop/rollback procedure.

9 | Reassessment, suspension, and revocation

An ALLOW is bounded in site, asset, actor, scope, time and conditions. It is not perpetual. Material changes to ventilation, occupancy, target contaminant, equipment configuration, firmware, maintenance state, measured byproducts, governing rules, or authority relationship require a new examination.

An emergency stop may be executed under a preauthorized safety protocol; the stop and resulting evidence must be recorded. A change in evidence does not silently rewrite a prior decision: issue a new version with supersession, reason, responsible authority and time.

10 | Decision record schema (minimum)

Field	Required content
Identity	Proposal ID, revision, site, asset, actor, decision time, evidence cut-off
Claim	Specific endpoint, population, comparator, scope and prohibited inferences
Evidence	Immutable/raw references, provenance, methods, uncertainty, transfer matrix
Authority/standing	Rule or permission, actor, capacity, standing basis, authority relationship
Gate outcomes	G1–G6 finding, missing evidence, reviewer, rationale
Disposition	ALLOW/HOLD/DENY/ESCALATE, action boundary, conditions and expiry
Outcome	Baseline, observations, adjustments, deviations, incidents, stop events
Review	Reopen criteria, reassessment triggers, superseding decision link

11 | Illustrative decision record

Proposal: AIR-DEPLOY-EXAMPLE-001 v0.2 | Setting: occupied classroom | Evidence: lab surrogate report (unverified illustrative reference), field baseline missing | Claim A: reduced infection risk | Claim A disposition: HOLD | Deployment B disposition: HOLD | Standing: not yet documented | Release: field-relevance study, safety review, documented site permission, predeclared monitoring and stop rules | Reopen: new admissible evidence and competent authorizations.

This is not a determination about any named technology, school, manufacturer, EPA project or Blueprint project.

12 | Institutional exchange and potential work packages

Examine: nonconfidential case review, evidence gap matrix, claims and decision boundaries. Establish: jointly specified evidence schema, standing records, transfer criteria, authorization gates and decision templates. Operate: versioned decision and outcome records, periodic reassessment and auditability. These are possible discussion lanes, not approved, contracted or funded CRADA work.

Recommended initial written exchange: ask the scientific team for one nonconfidential use case, target efficacy claim, lab protocol, proposed field context, safety/regulatory interface, accountable decision owner, and intended consequence. TA14 can respond with a bounded examination matrix. Confidential information, intellectual property, data access, paid work, and joint development require separate written agreements.

13 | Limitations, independence, and responsible claims

This framework is a governance examination aid, not a scientific validation, engineering design, legal opinion, risk assessment, certification, or regulatory authorization. A complete governance record does not establish a device is safe or effective. No EPA or Blueprint endorsement, affiliation, funding, data access or collaboration is claimed. The research-to-deployment challenge areas reflect the correspondence described to TA14; all detailed technical examination constructs in this edition are TA14 proposals for discussion.

14 | Eight-image showroom teaching sequence

Frame	Teaching focus
01	Laboratory efficacy is not field authorization
02	Precisely bounded claims and quantitative evidence
03	Provenance, raw observations and baseline continuity
04	Scientific transferability matrix and six gates
05	Human consequence, byproducts and stop conditions
06	Applicable authority and four-anchor standing
07	Separate claim and deployment dispositions
08	Authorized execution, outcome and reassessment

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